



Data Mining

HW3: News Recommendation Prediction

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Task Description

Topic: News Recommendation Prediction

You are given several entries of user interaction history, you need to use these information to predict whether a user will click on a news or not.

Goal: Use any model or method to predict whether a user will click on a news.

1. Submit a report and source code to E3
2. Upload your prediction to Kaggle competition





Dataset

You are given 2 datasets (download via Kaggle), **train** and **test**, each with 3 files:

- behaviors.tsv
- entity_embedding.vec
- news.tsv



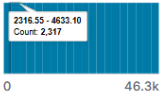
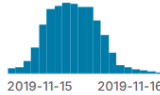
Dataset (cont.)

- behaviors.tsv
 - **user_id**: user id.
 - **time**: the time the user visited the website homepage.
 - **clicked_news**: **space-separated list** of user's click history.
 - **impressions**: **space-separated list (15 items)**, whether user clicked on the recommended news or not. (Format: {news_id}-{clicked}, clicked is 0 or 1.)

Dataset (cont.) - behaviors.tsv -train

1	2	3	4	5
id	user_id	time	clicked_news	impressions
0	U1349561	11/11/2019 9:31:08 AM	N410559 N109405 N79284 N812877 N311012 N362	N383574-0 N727666-0 N169045-0 N846428-0 N703824-0 N2970-0 N751592-0 N159567-0 N
1	U2788121	11/9/2019 9:13:19 AM	N642952 N253717 N857922 N684266 N291776 N12	N184823-0 N107900-0 N29500-0 N122187-0 N487132-0 N811701-0 N49569-0 N460476-0 N
2	U686145	11/12/2019 6:21:28 AM	N900496 N118253 N510477 N167498 N693772	N499466-0 N665940-0 N394508-1 N386423-0 N396755-0 N88097-0 N489400-0 N624080-0
3	U2794941	11/13/2019 9:30:05 AM	N417895	N96322-0 N909778-0 N656987-0 N594694-0 N306259-1 N820689-0 N602779-0 N374313-0
4	U1838845	11/10/2019 5:03:16 AM	N187833 N272183 N482344 N65242 N211696 N194	N27904-0 N731054-0 N281766-0 N177459-0 N768567-0 N530889-0 N837454-0 N647236-0
5	U137749	11/9/2019 4:06:02 PM	N522916	N537665-0 N367040-0 N219690-0 N407682-0 N503484-0 N627384-0 N103931-0 N225458-0
6	U1957641	11/14/2019 11:34:50 PM	N271189 N187833 N232675 N746132 N7058 N8525	N577516-0 N453490-1 N910527-0 N830356-0 N30641-0 N520802-0 N618298-0 N860603-0
7	U17537	11/13/2019 9:08:41 AM	N436802 N93956 N24040 N501650 N315863 N4419	N668670-0 N619299-0 N71990-0 N508258-0 N88762-0 N656987-0 N341483-0 N894847-0 N
8	U436289	11/9/2019 10:47:35 AM	N716914 N887847 N603640 N394312	N370435-0 N460476-1 N302297-0 N780845-0 N103931-0 N48050-0 N44046-0 N534683-0 N
9	U1166697	11/12/2019 10:23:09 AM	N188540 N520977 N712420 N536321 N719196 N40	N859728-0 N379668-0 N272309-0 N447659-0 N291965-0 N866063-0 N767489-1 N96322-0
10	U2617573	11/11/2019 11:37:41 AM	N117280 N324172 N794824 N284391 N279442 N22	N796182-1 N479929-0 N69540-0 N751592-0 N881610-0 N820815-0 N386423-0 N183402-0
11	U1714313	11/14/2019 7:50:07 PM	N266681 N446056 N871166 N610647 N272183 N41	N54091-0 N25356-0 N380571-1 N678701-0 N202022-0 N483394-0 N592013-0 N415011-0 N
12	U768033	11/13/2019 5:13:51 AM	N366900 N50633 N407535 N407535	N677098-0 N84275-0 N567870-0 N661999-0 N457186-0 N324900-0 N909778-1 N249559-0
13	U437037	11/11/2019 11:21:30 AM	N118253 N228510 N709634 N247291 N99675 N274	N772382-0 N894693-0 N614980-0 N735926-1 N563586-0 N374607-1 N258330-0 N589164-0
14	U2195893	11/14/2019 6:38:23 AM	N290383 N283117 N863760 N54189 N9767 N50109	N821627-0 N32531-1 N290656-0 N789280-0 N756975-0 N515321-0 N501377-0 N392324-0
15	U2384397	11/13/2019 4:32:24 AM	N247529 N76029 N37130 N7891 N5350 N614294 N	N71759-0 N324900-0 N567870-0 N194161-1 N909778-0 N96322-0 N36150-0 N791814-0 N8
16	U2650561	11/10/2019 4:15:22 PM	N136796	N97799-0 N839519-0 N315863-0 N750332-0 N184571-0 N270741-0 N114816-0 N545932-0
17	U760757	11/13/2019 2:42:03 PM	N342792 N900496 N77849 N298916 N391316 N388	N906565-0 N161534-0 N13456-0 N536734-0 N872734-1 N307869-0 N429893-0 N668670-0

Dataset (cont.) - behaviors.tsv - test

id	user_id	time	clicked_news	impressions
 2316.35 - 4633.10 Count: 2,317 0 46.3k	43296 unique values	 2019-11-15 2019-11-16	43140 unique values	46332 unique values
0	U718485	11/15/2019 9:54:29 AM	N63961 N291636 N631290 N676230 N783386 N32517 N804008 N880966 N414710 N783386 N613489 N394466 N52664...	N447295 N493208 N164159 N142788 N403902 N276971 N149487 N66292 N736892 N497002 N886867 N735198 N8406...
1	U1816521	11/15/2019 11:32:28 AM	N376420 N588093 N800305 N173161 N708836 N512717 N219963 N441051 N511261 N433106 N431979 N43661 N2361...	N732503 N311166 N54091 N489561 N130356 N640537 N804645 N127801 N565385 N395950 N149487 N145609 N6122...
2	U52997	11/15/2019 5:38:33 AM	N551819 N889275 N594708 N89735 N352039 N855843 N733154	N30641 N216680 N860603 N415074 N489561 N186664 N285189 N303522 N510834 N298538 N639144 N88020 N56538...
3	U1492017	11/15/2019 9:46:54 AM	N595387 N662468 N108292 N604249 N305090 N504730 N386479 N301604	N628343 N371121 N204115 N650330 N647327 N860603 N186916 N211437 N840618 N642154 N128487 N380571 N985...
4	U000000	11/15/2019 7:00:00 AM	N000001 N000001	N000000 N000000



Dataset (cont.)

- `news.tsv`
 - `news_id`
 - `category`: news category.
 - `subcategory`: more fine-grained news category.
 - `title`: title of the news.
 - `abstract`: (possibly empty) abstract of the news.
 - `URL`



Dataset (cont.)

- `news.tsv` (cont.)
 - **title_entities**: special entities in news title that is linked with [Wikidata](#). (Format: **json list**, each item is a linked entity)
 - **abstract_entities**: special entities in news abstract.

Dataset (cont.) - news .tsv - train

news_id	category	subcategory	title	abstract	URL	title_entities	abstract_entities
101527 unique values	sports 32% news 30% Other (39029) 38%	newsus 14% football_nfl 12% Other (75247) 74%	98388 unique values	[null] 5% What's the weathe... 0% Other (95677) 94%	101526 unique values	[] 26% [{"Label": "Housto... 0% Other (75409) 74%	[] [{"Label": "Power... Other (75985)
N621266	lifestyle	lifestyleroys	The Brands Queen Elizabeth, Prince Charles, and Prince Philip Swear By	Shop the notebooks, jackets, and more that the royals can't live without.	https://assets.msn.com/labs/mind/AAGH0ET.html	[{"Label": "Prince Philip, Duke of Edinburgh", "Type": "P", "WikidataId": "Q80976", "Confidence": 1.0...]	[]
N318047	news	newsscienceandtechno logy	Walmart Slashes Prices on Last-Generation iPads	Apple's new iPad releases bring big deals on last year's models.	https://assets.msn.com/labs/mind/AABmf2I.html	[{"Label": "iPad", "Type": "J", "WikidataId": "Q2796", "Confidence": 0.999, "OccurrenceOffsets": 42...]	[{"Label": "iPad", "Type": "J", "WikidataId": "Q2796", "Confidence": 0.9, "OccurrenceOffset": 12...]
N162003	health	weightloss	50 Worst Habits For Belly Fat	These seemingly harmless habits are holding you back and keeping you from shedding that unwanted bel...	https://assets.msn.com/labs/mind/AAB19MK.html	[{"Label": "Adipose tissue", "Type": "C", "WikidataId": "Q193583", "Confidence": 1.0, "OccurrenceOff...]	[{"Label": "Adipo tissue", "Type": "C", "WikidataId": "Q193583", "Confidence": 1.0, "OccurrenceOff...]

Dataset (cont.) - news . tsv - test

news_id	category	subcategory	title	abstract	URL	title_entities	abstract_entities
72023 unique values	news 30% sports 30% Other (28618) 40%	newsus 14% football_nfl 12% Other (53691) 75%	70316 unique values	[null] 5% What's the weathe... 0% Other (68176) 95%	72022 unique values	[] 26% [{"Label": "Housto... 0% Other (53242) 74%	[] [{"Label": "Power... Other (53306)
N621266	lifestyle	lifestyleroyals	The Brands Queen Elizabeth, Prince Charles, and Prince Philip Swear By	Shop the notebooks, jackets, and more that the royals can't live without.	https://assets.msn.com/labs/mind/AAGH0ET.html	[{"Label": "Prince Philip, Duke of Edinburgh", "Type": "P", "WikidataId": "Q80976", "Confidence": 1....	[]
N162003	health	weightloss	50 Worst Habits For Belly Fat	These seemingly harmless habits are holding you back and keeping you from shedding that unwanted bel...	https://assets.msn.com/labs/mind/AAB19MK.html	[{"Label": "Adipose tissue", "Type": "C", "WikidataId": "Q193583", "Confidence": 1.0, "OccurrenceOff...	[{"Label": "Adipo tissue", "Type": "C", "WikidataId": "Q193583", "Confidence": 1.0, "OccurrenceOff...
N603780	health	medical	Dispose of unwanted prescription drugs during the DEA's Take Back Day		https://assets.msn.com/labs/mind/AAISxPN.html	[{"Label": "Drug Enforcement Administration", "Type": "O", "WikidataId": "Q622899", "Confidence": 0....	[]



Dataset (cont.)

- `entity_embedding.vec`
 - For convenience, we provide **100-D** trained embeddings of special entities. (tab separated)
 - **You can also use your preferred embeddings.**

Dataset (cont.) - entity_embedding.vec - train

Q34433	0.017808	-0.073256	0.102521	-0.059926	-0.060665	0.027027	-0.091728	-0.003057	-0.170
Q41	-0.063388	-0.181451	0.057501	-0.091254	-0.076217	-0.052525	0.050500	-0.224871	-0.018
Q56037	0.021550	-0.044888	-0.027872	-0.128843	0.066651	-0.072159	0.019879	-0.183956	0.0806
Q1860	0.060958	0.069934	0.015832	0.079471	-0.023362	-0.125007	-0.043618	0.134063	-0.121
Q7737	-0.021237	0.176011	-0.078886	0.041470	0.136488	-0.063177	-0.013134	-0.048977	-0.072
Q39631	-0.093106	-0.052002	0.020556	-0.020801	0.043180	-0.072321	0.000910	0.028156	0.1763
Q30	-0.115737	-0.179113	0.102739	-0.112469	-0.101853	-0.177516	0.015860	-0.092626	0.0867
Q60	-0.051036	-0.165637	0.132802	-0.089949	-0.146637	-0.142246	0.103853	-0.129651	0.0962
Q847017	-0.043970	-0.085714	0.011526	0.022439	0.126266	-0.109745	-0.091008	0.047532	0.0080
Q183	0.052780	-0.139523	-0.027571	-0.196823	0.059276	-0.109428	-0.000309	-0.136393	0.1027
Q2736	-0.091826	-0.021255	-0.049415	-0.167199	0.028283	-0.010620	-0.026264	-0.082447	0.0893
Q21198	-0.096089	-0.006838	-0.027840	-0.098632	0.020599	0.047792	-0.020709	-0.056511	0.0352
Q131524	0.004348	0.028879	-0.008639	-0.080811	0.078301	-0.080898	-0.197224	-0.032734	0.0285
Q12788174	-0.080052	0.111832	0.076317	-0.084863	-0.021116	0.063792	-0.153752	-0.013421	
Q142	0.044556	0.005308	0.135917	-0.203567	0.024435	-0.098517	-0.039502	-0.222709	0.0901
Q3561	-0.049529	0.010307	0.126421	0.011403	0.042286	-0.161760	0.074351	-0.150197	0.0184
Q43845	-0.022434	-0.040100	-0.016155	-0.053598	0.061538	-0.217902	-0.178407	0.022501	0.1604
Q298	0.013886	-0.088592	-0.004194	-0.072603	0.001292	-0.045497	0.066282	-0.035162	0.1502
Q2887	0.039071	-0.076391	0.015528	0.011815	-0.022475	-0.039344	0.104316	0.002287	0.1305
Q1321	-0.033441	0.103234	-0.070237	0.123462	0.098123	-0.017173	-0.013411	0.002179	-0.095

Dataset (cont.) - entity_embedding.vec - test

Q34433	0.017808	-0.073256	0.102521	-0.059926	-0.060665	0.027027	-0.091728	-0.003057	-0.170798	0.111819
Q41	-0.063388	-0.181451	0.057501	-0.091254	-0.076217	-0.052525	0.050500	-0.224871	-0.018145	0.030722
Q56037	0.021550	-0.044888	-0.027872	-0.128843	0.066651	-0.072159	0.019879	-0.183956	0.080640	0.069166
Q1860	0.060958	0.069934	0.015832	0.079471	-0.023362	-0.125007	-0.043618	0.134063	-0.121691	0.089166
Q39631	-0.093106	-0.052002	0.020556	-0.020801	0.043180	-0.072321	0.000910	0.028156	0.176303	0.035396
Q30	-0.115737	-0.179113	0.102739	-0.112469	-0.101853	-0.177516	0.015860	-0.092626	0.086708	0.057850
Q60	-0.051036	-0.165637	0.132802	-0.089949	-0.146637	-0.142246	0.103853	-0.129651	0.096265	0.017288
Q183	0.052780	-0.139523	-0.027571	-0.196823	0.059276	-0.109428	-0.000309	-0.136393	0.102720	0.121319
Q2736	-0.091826	-0.021255	-0.049415	-0.167199	0.028283	-0.010620	-0.026264	-0.082447	0.089347	0.167264
Q21198	-0.096089	-0.006838	-0.027840	-0.098632	0.020599	0.047792	-0.020709	-0.056511	0.035283	0.178778
Q131524	0.004348	0.028879	-0.008639	-0.080811	0.078301	-0.080898	-0.197224	-0.032734	0.028527	0.158976
Q12788174	-0.080052	0.111832	0.076317	-0.084863	-0.021116	0.063792	-0.153752	-0.013421	-0.073218	
Q142	0.044556	0.005308	0.135917	-0.203567	0.024435	-0.098517	-0.039502	-0.222709	0.090129	0.066471
Q3561	-0.049529	0.010307	0.126421	0.011403	0.042286	-0.161760	0.074351	-0.150197	0.018418	0.146195
Q43845	-0.022434	-0.040100	-0.016155	-0.053598	0.061538	-0.217902	-0.178407	0.022501	0.160479	0.136865
Q298	0.013886	-0.088592	-0.004194	-0.072603	0.001292	-0.045497	0.066282	-0.035162	0.150286	0.127842
Q2887	0.039071	-0.076391	0.015528	0.011815	-0.022475	-0.039344	0.104316	0.002287	0.130517	0.079433
Q1321	-0.033441	0.103234	-0.070237	0.123462	0.098123	-0.017173	-0.013411	0.002179	-0.095182	0.177995
Q155	0.034837	-0.184234	0.050972	-0.070311	-0.033401	-0.154621	0.054434	-0.090577	0.124591	0.172820
Q5372	-0.109987	-0.210057	-0.052240	-0.149397	-0.165659	0.148978	-0.014366	-0.141065	0.024355	0.097681
Q15180	-0.113812	0.003113	-0.007003	-0.082540	0.081815	-0.159194	-0.070831	-0.181713	0.122815	0.068568
Q408	0.030718	-0.167667	-0.032142	-0.097031	-0.009249	-0.153150	0.054930	-0.050344	0.089500	0.072373



Kaggle

- [Competition Link](#)
- Create a team with your **student ID**, we use this information for grading.
- If you failed to do so under any circumstances, there will be **a penalty of 5 points** to your score, so be sure to use the correct team name.



Kaggle (cont.)

- Public leaderboard is calculated with 50% of the test set, private leaderboard is calculated with the other 50%, the final standings may be different.
- Therefore, **please DO NOT overtune your model to fit the public leaderboard, or you will suffer from overfitting.**



Kaggle (cont.)

- The scoring metric is **Area Under the Curve of ROC (AUC ROC)**. If you don't know what it is, [here](#) is a quick video for you.
- We have set a simple baseline and a strong baseline, beat them to get higher score.
- You can submit at most 10 times each day and choose 2 of the submissions to be scored for the private leaderboard, or will otherwise default to the best public scoring submissions.



Kaggle Submission Format

- Report user with their **index (column “id” in test_behaviors.tsv, not “user_id”)**, and the corresponding prediction to each news.
- There should be **46333 x 16** entries in your csv file, **with columns “id” and “p1 - p15”**.
- The order of the ids does not matter. Refer to [sample_submission.csv](#) for the correct format.



Kaggle Submission Format

- The “p1 - p15” columns should contain the probability predicted by your model, of the user clicking the 1st news, to the 15th news.
- Although we ask you to provide the “probability”, the value does not have to be in $[0, 1]$, because the AUC metric still works without that rule.



Baseline Approach

- **Disclaimer:** The following approach are just for your reference. You are not required to follow any of them. We encourage you to invent new ways to solve this problem.

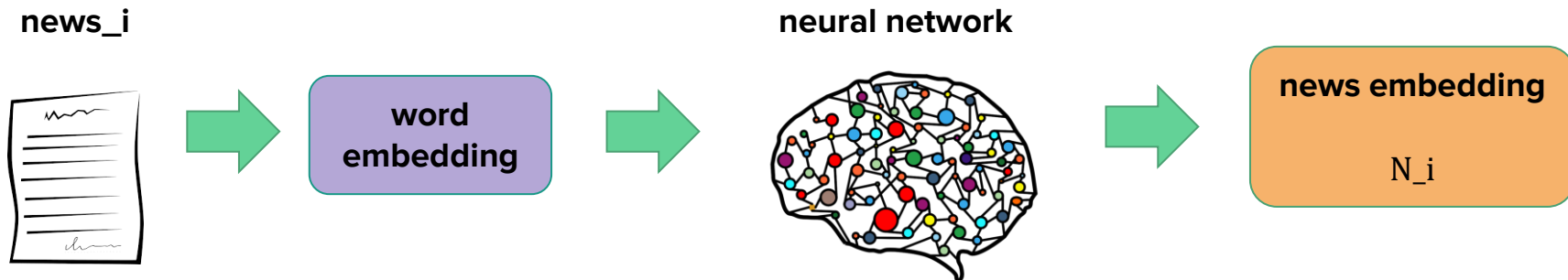


Baseline Approach

- The amount of data is huge, and we need to predict 15 news for each user. The news itself is a short (possibly long) paragraph, we won't be able to fit them all in a single model. Therefore, try to find a way to **encode each news as a "news embedding"**. This will also help us comparing the similarity between each news.

Baseline Approach

- To build the news embedding, you can use BERT as the encoder, or train a neural network using your own preferred word embedding (e.g. GloVe, flair).



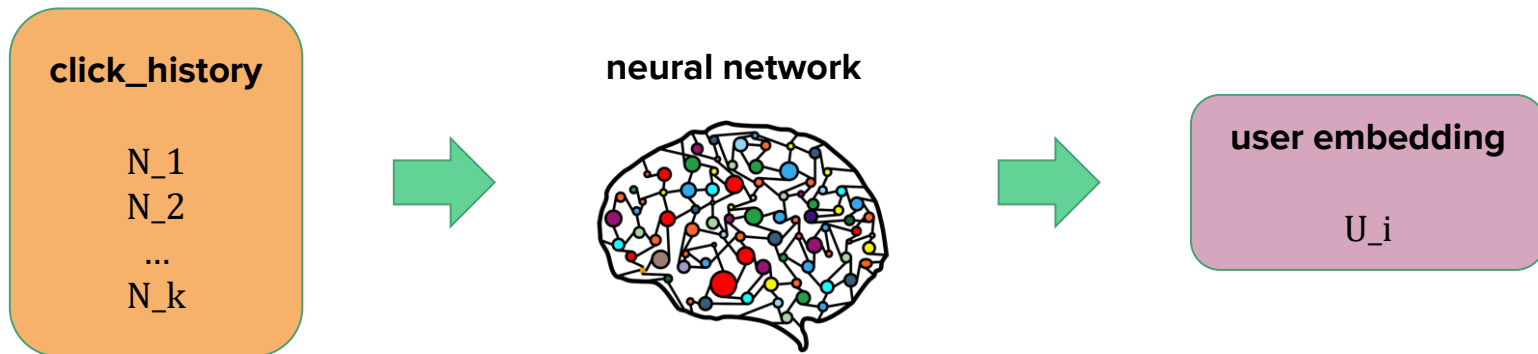


Baseline Approach

- For each user, we need to find news that are to their liking. To determine whether a user will like a news or not, we shall build **“user embedding”**, and then compare this embedding with the “news embedding” we just got.

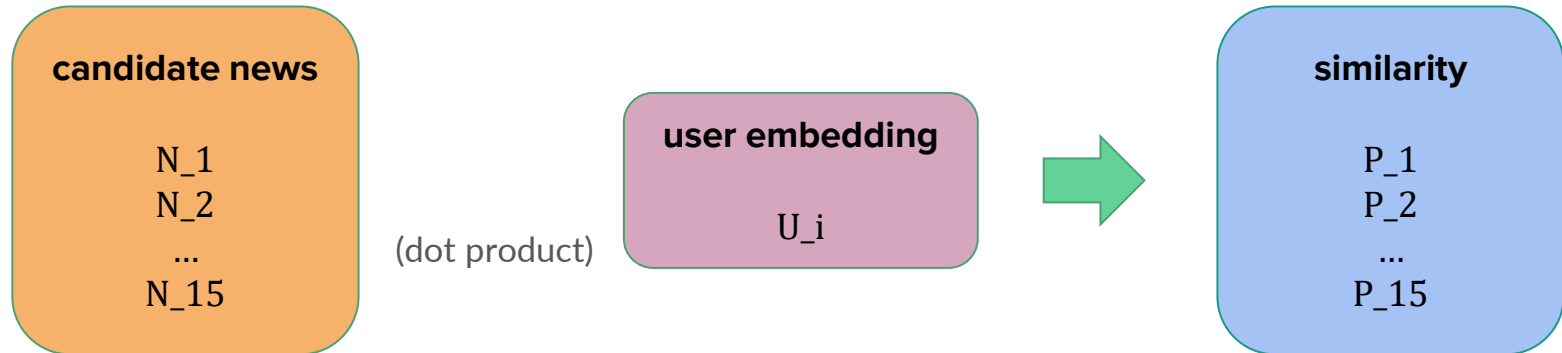
Baseline Approach

- One easy way to build a user embedding is to consider their click history, and use a model to train the embedding.



Baseline Approach

- Finally, to calculate the similarity, we can use **dot product** to compare the user embedding with each news embedding.





Tips

- There are a lot of additional information in the dataset, you can find a way to use them in your approach.
- LLMs may not help you predicting the news in this task, since the data is too big. However, since LLMs can receive huge inputs, you can leverage this to generate “enriched” description of user history, which may help your model predict better.



Report

Please answer the following questions, provide your thinking process as detailed as possible:

1. (10%) Describe how you solve this problem. Details include preprocessing, embeddings, model selection, hyperparameters should be provided.

(There will be penalties to your score if you failed to do so)



Report (cont.)

2. (5%) Choose a **variable (e.g. different model, different approach)** excluding hyperparameters and compare their performance.
(You cannot change ONLY the hyperparameters)

(5%) Explain what causes the difference of performance or why.

(For example, you were using BERT as the encoder, in Q2 you tried RoBERTa and found that using RoBERTa will yield better performance. Give a reason why this is the case)



Report (cont.)

3. (10%) Do **error analysis** or **case study**. Is there anything worth mentioning while checking the mispredicted data? Share with us. Anytime you try to make a conclusion about the data or model, you should provide concrete data example.

[e.g. I think the model predicts poorly when ... (provide error examples)]

There is no “correct answer” to above questions, just do your best and answer them in detail.



Grading Policy

- **Kaggle (70% of total)**, details in the next slides.
- **Report (30% of total)**, points are attached at the start of each question.

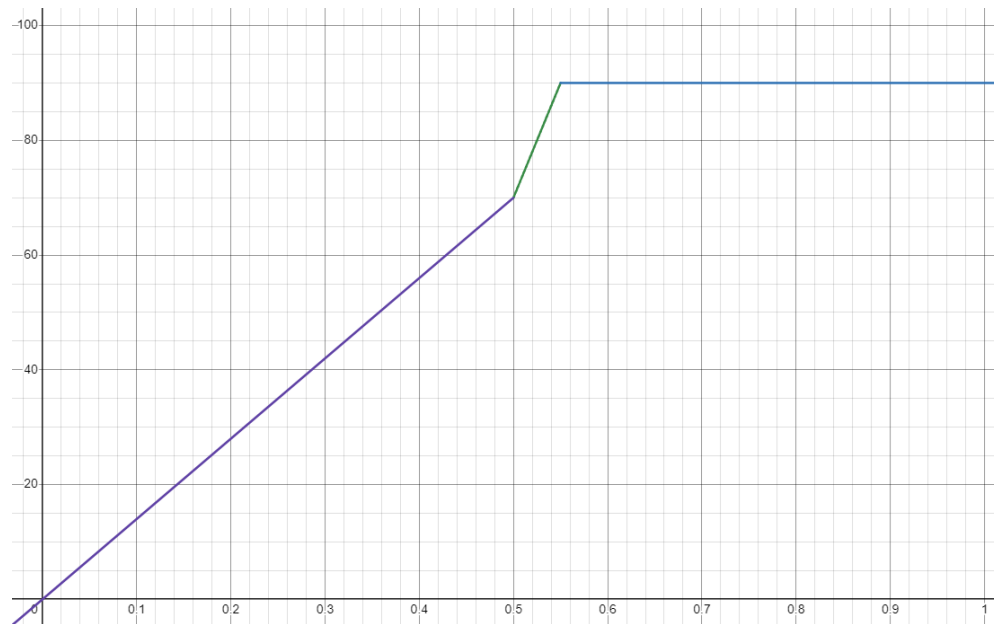


Grading Policy - Kaggle

- **Baselines (90% of Kaggle; 30% public, 70% private)**
 - We have 2 baselines: Strong Baseline (**B_1**), Simple Baseline (**B_2**)
 - **Your score** > **B_1**: 90
 - **B_1** > **Your score** (X) > **B_2**: $(X - B_2) / (B_1 - B_2) * 20 + 70$.
 - **Your score** (X) < **B_2**: $X / B_2 * 70$.

Grading Policy - Kaggle

- Assume that:
 - $B_1 = 0.55$
 - $B_2 = 0.5$
- x-axis: F1
- y-axis: score





Grading Policy - Kaggle

- **Ranking (10% of Kaggle)**
 - Compete with your classmates.
 - Rank 1-5: 10%
 - Rank 6-10: 8%
 - ...
 - Rank 21-25: 2%
 - Rank > 25: no points :(



E3 Submission

- Submit your source code and report to E3 before **6/6(Fri.) 23:59**, no late submissions will be accepted.
- Please submit your source code in **python source (.py)**. For jupyter notebooks, you can use the export function to obtain the executable script. **Violation of this rule will cause a deduction of 5 points to your grade.**
- **Do NOT put full code in your report.** Only short code snippets and pseudocode (for demonstration purposes) are allowed, and they should be properly formatted.



E3 Submission

Submission format:

- hw3_<student_id>.zip
 - **source code:** hw3_<student_id>.py or other library files (.py) you made (**python files only!**)
 - **report:** hw3_<student_id>.pdf



Contact

- If you have any questions about Final Project, please feel free to contact with TA by email:
 - Hank Chen (bwchen@cs.nycu.edu.tw)
- **Kindly format your email** so that I can reply it ASAP.
 - **Prefix your email title with [DM]**. (e.g. [DM] Questions about Final Project)
 - Attach your student ID if necessary.